

OpenStax Calculus Volume 3

Section 6.7 Exercises: Stokes' Theorem

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For the following exercises, without using Stokes' theorem, calculate directly both the flux of $\text{curl } \mathbf{F} \cdot \mathbf{N}$ over the given surface and the circulation integral around its boundary, assuming all boundaries have positive orientation.

1. $\mathbf{F}(x, y, z) = y^2\mathbf{i} + z^2\mathbf{j} + x^2\mathbf{k}$; S is the first-octant portion of plane $x + y + z = 1$.
2. $\mathbf{F}(x, y, z) = z\mathbf{i} + x\mathbf{j} + y\mathbf{k}$; S is hemisphere $z = (a^2 - x^2 - y^2)^{1/2}$.
3. $\mathbf{F}(x, y, z) = y^2\mathbf{i} + 2x\mathbf{j} + 5\mathbf{k}$; S is hemisphere $z = (4 - x^2 - y^2)^{1/2}$.
4. $\mathbf{F}(x, y, z) = z\mathbf{i} + 2x\mathbf{j} + 3y\mathbf{k}$; S is upper hemisphere $z = \sqrt{9 - x^2 - y^2}$.
5. $\mathbf{F}(x, y, z) = (x + 2z)\mathbf{i} + (y - x)\mathbf{j} + (z - y)\mathbf{k}$; S is a triangular region with vertices $(3, 0, 0)$, $(0, 3/2, 0)$, and $(0, 0, 3)$.
6. $\mathbf{F}(x, y, z) = 2y\mathbf{i} - 6z\mathbf{j} + 3x\mathbf{k}$; S is a portion of paraboloid $z = 4 - x^2 - y^2$ and is above the xy-plane.

For the following exercises, use Stokes' theorem to evaluate $\iint_S (\text{curl } \mathbf{F} \cdot \mathbf{N}) dS$ for the vector fields and surface.

7. $\mathbf{F}(x, y, z) = xy\mathbf{i} - z\mathbf{j}$ and S is the surface of the cube $0 \leq x \leq 1, 0 \leq y \leq 1, 0 \leq z \leq 1$, except for the face where $z = 0$, and using the outward unit normal vector.
8. $\mathbf{F}(x, y, z) = xy\mathbf{i} + x^2\mathbf{j} + z^2\mathbf{k}$; and S is the part of paraboloid $z = x^2 + y^2$ below plane $z = y$, and using the outward normal vector.
9. $\mathbf{F}(x, y, z) = 4y\mathbf{i} + z\mathbf{j} + 2y\mathbf{k}$ and S is the part of sphere $x^2 + y^2 + z^2 = 4$ above plane $z = 0$, and using the outward normal vector.
10. Use Stokes' theorem to evaluate $\int_C [2xy^2z dx + 2x^2yz dy + (x^2y^2 - 2z) dz]$, where C is the curve given by $x = \cos t, y = \sin t, z = \sin t, 0 \leq t \leq 2\pi$, traversed in the direction of increasing t.
[Figure: A vector field in three dimensional space. The arrows are larger the further they are from the x, y plane. The arrows curve up from below the x, y plane and slightly above it. The rest tend to curve down and horizontally. An oval-shaped curve is drawn through the middle of the space.]

11. [T] Use a computer algebra system (CAS) and Stokes' theorem to approximate line integral $\int_C (y dx + z dy + x dz)$, where C is the intersection of plane $x + y = 2$ and surface $x^2 + y^2 + z^2 = 2(x + y)$, traversed counterclockwise viewed from the origin.
12. [T] Use a CAS and Stokes' theorem to approximate line integral $\int_C (3y dx + 2z dy - 5x dz)$, where C is the intersection of the xy-plane and hemisphere $z = \sqrt{1 - x^2 - y^2}$, traversed counterclockwise viewed from the top—that is, from the positive z-axis toward the xy-plane.
13. [T] Use a CAS and Stokes' theorem to approximate line integral $\int_C [(1 + y)z dx + (1 + z)xy dy + (1 + x)yz dz]$, where C is a triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ oriented counterclockwise.
14. Use Stokes' theorem to evaluate $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = e^{xy} \cos z \mathbf{i} + x^2 z \mathbf{j} + xy \mathbf{k}$, and S is half of sphere $x = \sqrt{1 - y^2 - z^2}$, oriented out toward the positive x-axis.
15. [T] Use a CAS and Stokes' theorem to evaluate $\iint_S (\text{curl } \mathbf{F} \cdot \mathbf{N}) dS$, where $\mathbf{F}(x, y, z) = x^2 y \mathbf{i} + xy^2 \mathbf{j} + z^3 \mathbf{k}$ and S is the curve of the part of plane $3x + 2y + z = 6$ above cylinder $x^2 + y^2 = 4$, oriented clockwise when viewed from above.
16. [T] Use a CAS and Stokes' theorem to evaluate $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = (\sin(y + z) - yx^2 - \frac{y^3}{3}) \mathbf{i} + x \cos(y + z) \mathbf{j} + \cos(2y) \mathbf{k}$ and S consists of the top and the four sides but not the bottom of the cube with vertices $(\pm 1, \pm 1, \pm 1)$, oriented outward.
17. [T] Use a CAS and Stokes' theorem to evaluate $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = z^2 \mathbf{i} - 3xy \mathbf{j} + x^3 y^3 \mathbf{k}$ and S is the top part of $z = 5 - x^2 - y^2$ above plane $z = 1$, and S is oriented upward.
18. Use Stokes' theorem to evaluate $\iint_S (\text{curl } \mathbf{F} \cdot \mathbf{N}) dS$, where $\mathbf{F}(x, y, z) = z^2 \mathbf{i} + y^2 \mathbf{j} + x \mathbf{k}$ and S is a triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ with upward orientation.
19. Use Stokes' theorem to evaluate line integral $\int_C (z dx + x dy + y dz)$, where C is a triangle with vertices $(3, 0, 0)$, $(0, 0, 2)$, and $(0, 6, 0)$ traversed in the given order.
20. Use Stokes' theorem to evaluate $\int_C (\frac{1}{2} y^2 dx + z dy + x dz)$, where C is the curve of intersection of plane $x + z = 1$ and ellipsoid $x^2 + 2y^2 + z^2 = 1$, oriented clockwise from the origin.
[Figure: A diagram of an intersecting plane and ellipsoid in three dimensional space. There is an orange curve drawn to show the intersection.]
21. Use Stokes' theorem to evaluate $\iint_S (\text{curl } \mathbf{F} \cdot \mathbf{N}) dS$, where $\mathbf{F}(x, y, z) = x \mathbf{i} + y^2 \mathbf{j} + ze^{xy} \mathbf{k}$ and S is the part of surface $z = 1 - x^2 - 2y^2$ with $z \geq 0$, oriented upward.
22. Use Stokes' theorem to evaluate $\iint_S (\text{curl } \mathbf{F} \cdot \mathbf{N}) dS$, for vector field $\mathbf{F}(x, y, z) = z \mathbf{i} + 3x \mathbf{j} + 2z \mathbf{k}$ where S is surface $z = 1 - x^2 - y^2, z \geq 0$, C is boundary circle $x^2 + y^2 = 1$, and S is oriented in the positive z-direction.
23. Use Stokes' theorem to evaluate $\iint_S (\text{curl } \mathbf{F} \cdot \mathbf{N}) dS$, for vector field $\mathbf{F}(x, y, z) = -\frac{3}{2} y^2 \mathbf{i} - 2xy \mathbf{j} + yz \mathbf{k}$, where S is that part of the surface of plane $x + y + z = 1$ contained within triangle C with vertices $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$, traversed counterclockwise as viewed from above.
24. A certain closed path C in plane $2x + 2y + z = 1$ is known to project onto unit circle $x^2 + y^2 = 1$ in the xy-plane. Let c be a constant and let $\mathbf{R}(x, y, z) = x \mathbf{i} + y \mathbf{j} + z \mathbf{k}$. Use Stokes' theorem to evaluate $\int_C (c \mathbf{k} \times \mathbf{R}) \cdot d\mathbf{r}$.

25. Use Stokes' theorem and let C be the boundary of surface $z = x^2 + y^2$ with $0 \leq x \leq 2$ and $0 \leq y \leq 1$, oriented with upward facing normal. Define

$$\mathbf{F}(x, y, z) = [\sin(x^3) + xz]\mathbf{i} + (x - yz)\mathbf{j} + \cos(z^4)\mathbf{k} \text{ and evaluate } \int_C \mathbf{F} \cdot d\mathbf{r}.$$

26. Let S be hemisphere $x^2 + y^2 + z^2 = 4$ with $z \geq 0$, oriented upward. Let $\mathbf{F}(x, y, z) = x^2 e^{yz}\mathbf{i} + y^2 e^{xz}\mathbf{j} + z^2 e^{xy}\mathbf{k}$ be a vector field. Use Stokes' theorem to evaluate $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$.

27. Let $\mathbf{F}(x, y, z) = xy\mathbf{i} + (e^{z^2} + y)\mathbf{j} + (x + y)\mathbf{k}$ and let S be the graph of function $y = \frac{x^2}{9} + \frac{z^2}{9} - 1$ with $y \leq 0$ oriented so that the normal vector of S has a positive j component. Use Stokes' theorem to compute integral $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$.

28. Use Stokes' theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ where $\mathbf{F}(x, y, z) = y\mathbf{i} + z\mathbf{j} + x\mathbf{k}$ and C is a triangle with vertices $(0, 0, 0)$, $(2, 0, 0)$ and $(0, -2, 2)$ oriented counterclockwise when viewed from above.

29. Use the surface integral in Stokes' theorem to calculate the circulation of field F , $\mathbf{F}(x, y, z) = x^2 y^3 \mathbf{i} + \mathbf{j} + z\mathbf{k}$ around C , which is the intersection of cylinder $x^2 + y^2 = 4$ and hemisphere $x^2 + y^2 + z^2 = 16, z \geq 0$, oriented counterclockwise when viewed from above.

[Figure: A diagram in three dimensions of a vector field and the intersection of a cylinder and hemisphere. The arrows are horizontal and have negative x components for negative y components and have positive x components for positive y components. The curve of intersection between the hemisphere and cylinder is drawn in blue.]

30. Use Stokes' theorem to compute $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = \mathbf{i} + xy^2\mathbf{j} + xy^2\mathbf{k}$ and S is a part of plane $y + z = 2$ inside cylinder $x^2 + y^2 = 1$ and oriented upward.

[Figure: A diagram of a vector field in three dimensional space showing the intersection of a plane and a cylinder. The curve where the plane and cylinder intersect is drawn in blue.]

31. Use Stokes' theorem to evaluate $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = -y^2\mathbf{i} + x\mathbf{j} + z^2\mathbf{k}$ and S is the part of plane $x + y + z = 1$ in the first octant and oriented upward $x \geq 0, y \geq 0, z \geq 0$.

32. Let $\mathbf{F}(x, y, z) = xy\mathbf{i} + 2z\mathbf{j} - 2y\mathbf{k}$ and let C be the intersection of plane $x + z = 5$ and cylinder $x^2 + y^2 = 9$, which is oriented counterclockwise when viewed from the top. Compute the line integral of F over C using Stokes' theorem.

33. [T] Use a CAS and let $\mathbf{F}(x, y, z) = xy^2\mathbf{i} + (yz - x)\mathbf{j} + e^{xyz}\mathbf{k}$. Use Stokes' theorem to compute the surface integral of $\text{curl } F$ over surface S with inward orientation consisting of cube $[0, 1] \times [0, 1] \times [0, 1]$ with the right side missing.

34. Let S be ellipsoid $\frac{x^2}{4} + \frac{y^2}{9} + z^2 = 1$ oriented outward and let F be a vector field with component functions that have continuous partial derivatives. Compute $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$

35. Let S be the part of paraboloid $z = 9 - x^2 - y^2$ with $z \geq 0$ oriented upward. Verify Stokes' theorem for vector field $\mathbf{F}(x, y, z) = 3z\mathbf{i} + 4x\mathbf{j} + 2y\mathbf{k}$.

36. [T] Use a CAS and Stokes' theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ if $\mathbf{F}(x, y, z) = (3z - \sin x)\mathbf{i} + (x^2 + e^y)\mathbf{j} + (y^3 - \cos z)\mathbf{k}$, where C is the curve given by $x = \cos t, y = \sin t, z = 1, 0 \leq t \leq 2\pi$.

37. [T] Use a CAS and Stokes' theorem to evaluate $\iint_S \text{curl } \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = 2y\mathbf{i} + e^z\mathbf{j} - \arctan x\mathbf{k}$ with S as a portion of paraboloid $z = 4 - x^2 - y^2$ cut off by the xy -plane oriented upward.

38. [T] Use a CAS to evaluate $\iint_S \text{curl} \mathbf{F} \cdot d\mathbf{S}$, where $\mathbf{F}(x, y, z) = 2z\mathbf{i} + 3x\mathbf{j} + 5y\mathbf{k}$ and S is the surface defined parametrically by $\mathbf{r}(r, \theta) = r \cos \theta \mathbf{i} + r \sin \theta \mathbf{j} + (4 - r^2)\mathbf{k}$ ($0 \leq \theta \leq 2\pi$, $0 \leq r \leq 3$).
39. Let S be paraboloid $z = a(1 - x^2 - y^2)$, for $z \geq 0$, where $a > 0$ is a real number. Let $\mathbf{F} = \langle x - y, y + z, z - x \rangle$. For what value(s) of a (if any) does $\iint_S \text{curl} \mathbf{F} \cdot d\mathbf{S}$ have its maximum value?

For the following application exercises, the goal is to evaluate $A = \iint_S \text{curl} \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F} = \langle xz, -xz, xy \rangle$ and S is the upper half of ellipsoid $x^2 + y^2 + 8z^2 = 1$, where $z \geq 0$.

40. Evaluate a surface integral over a more convenient surface to find the value of A .
41. Evaluate A using a line integral.
42. Take paraboloid $z = x^2 + y^2$, for $0 \leq z \leq 4$, and slice it with plane $y = 0$. Let S be the surface that remains for $y \geq 0$, including the planar surface in the xz -plane. Let C be the semicircle and line segment that bounded the cap of S in plane $z = 4$ with counterclockwise orientation. Let $\mathbf{F} = \langle 2z + y, 2x + z, 2y + x \rangle$. Evaluate $\iint_S \text{curl} \mathbf{F} \cdot d\mathbf{S}$
[Figure: A diagram of a vector field in three dimensional space where a paraboloid with vertex at the origin, plane at $y=0$, and plane at $z=4$ intersect. The remaining surface is the half of a paraboloid under $z=4$ and above $y=0$.]

For the following exercises, let S be the disk enclosed by curve

$C: \mathbf{r}(t) = \langle \cos \phi \cos t, \sin t, \sin \phi \cos t \rangle$, for $0 \leq t \leq 2\pi$, where $0 \leq \phi \leq \frac{\pi}{2}$ is a fixed angle.

43. What is the length of C in terms of ϕ ?
44. What is the circulation of C of vector field $\mathbf{F} = \langle -y, -z, x \rangle$ as a function of ϕ ?
45. For what value of ϕ is the circulation a maximum?
46. Circle C in plane $x + y + z = 8$ has radius 4 and center $(2, 3, 3)$. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle 0, -z, 2y \rangle$, where C has a counterclockwise orientation when viewed from above.
47. *[Note: this exercise is reproduced verbatim from OpenStax; the closing instruction "compute the curl of $\mathbf{v}$ in a clockwise rotation" does not read as a well-formed mathematical task and may reflect an incomplete or malformed exercise in the source itself.]* Velocity field $\mathbf{v} = \langle 0, 1 - x^2, 0 \rangle$, for $|x| \leq 1$ and $|z| \leq 1$, represents a horizontal flow in the y -direction. Compute the curl of $\mathbf{v}$ in a clockwise rotation.
48. Evaluate integral $\iint_S \text{curl} \mathbf{F} \cdot d\mathbf{S}$ where $\mathbf{F} = -xz\mathbf{i} + yz\mathbf{j} + xye^z\mathbf{k}$ and S is the cap of paraboloid $z = 5 - x^2 - y^2$ above plane $z = 3$, and $\mathbf{n}$ points in the positive z -direction on S .

For the following exercises, use Stokes' theorem to find the circulation of the following vector fields around any smooth, simple closed curve C .

49. $\mathbf{F} = \nabla(x \sin ye^z)$
50. $\mathbf{F} = \langle y^2z^3, 2xyz^3, 3xy^2z^2 \rangle$